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(54) **SPLICING DISPLAY DEVICE**

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H10H 29/80 (2025.01)

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(58) **Field of Classification Search**

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See application file for complete search history.

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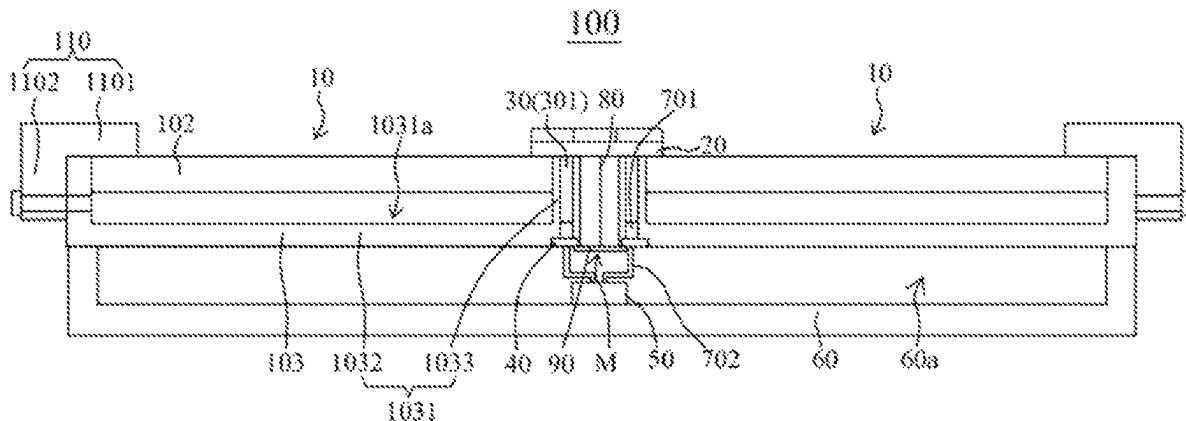
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(57) **ABSTRACT**

A splicing display device is provided. The splicing display device includes at least two first display units and a second display unit. The first display units are spliced and connected. Each of the first display units includes a non-display area. The second display unit is disposed on a light-emitting side of the first display units, and attached and fixed to a splicing position of the non-display areas of the two adjacent first display units.

20 Claims, 3 Drawing Sheets



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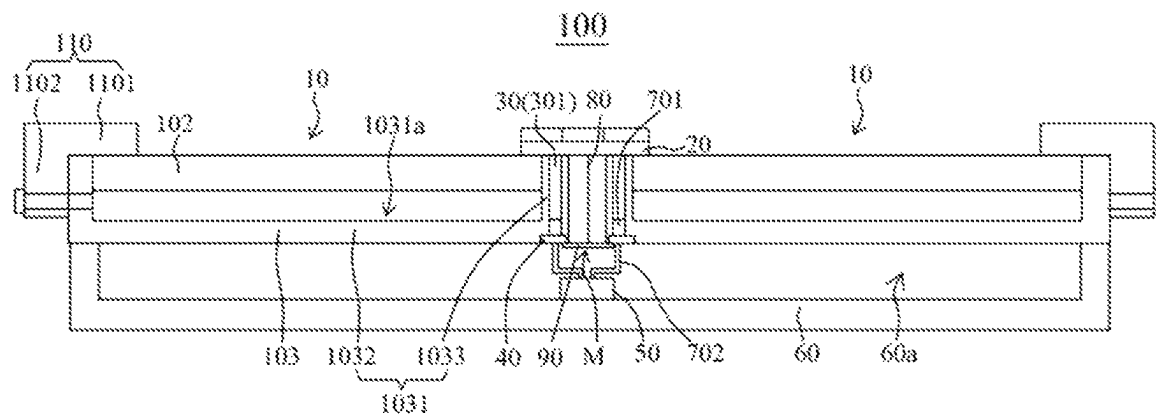


FIG. 1

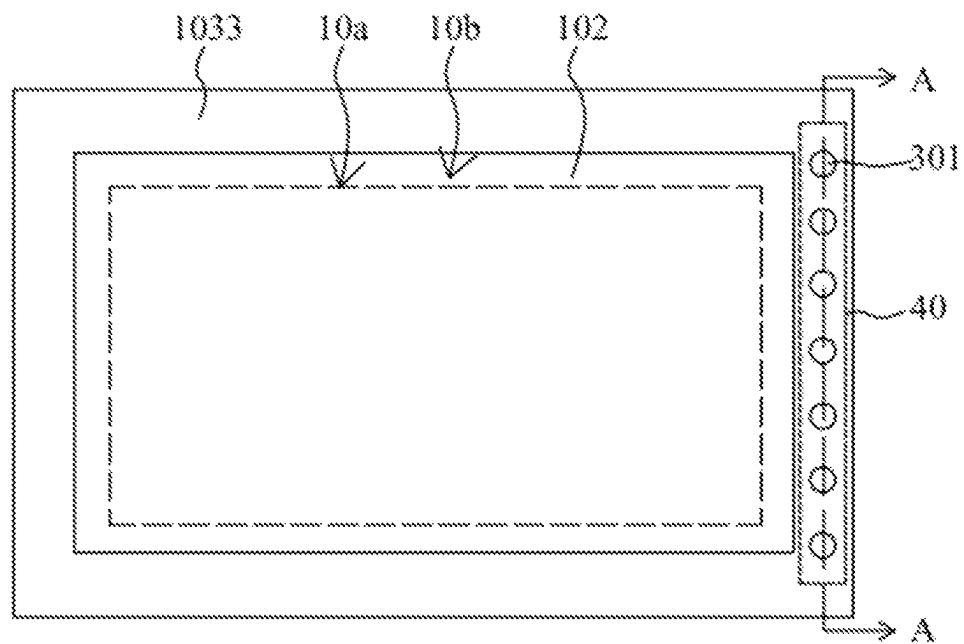


FIG. 2

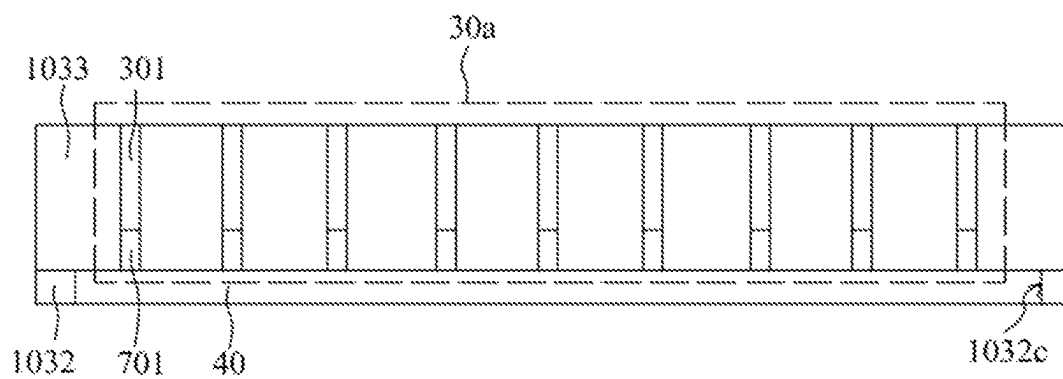


FIG. 3

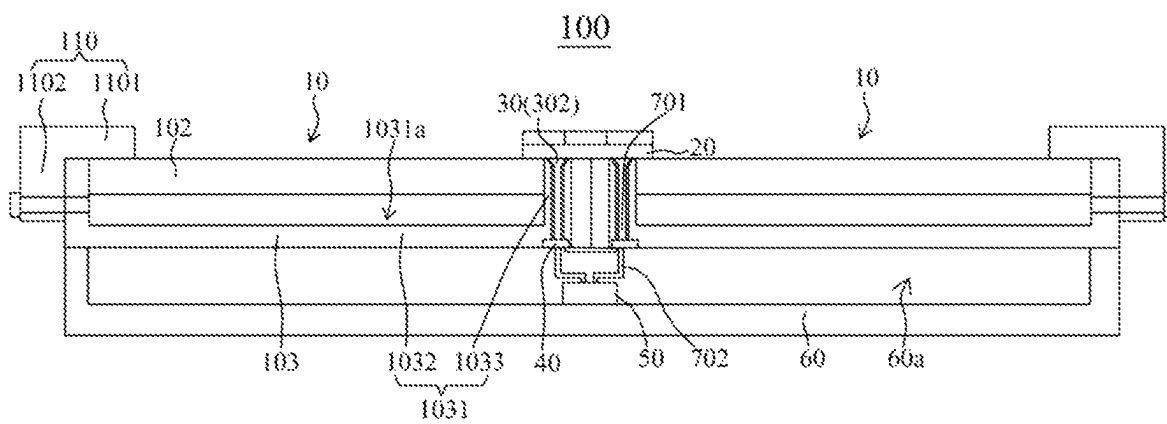


FIG. 4

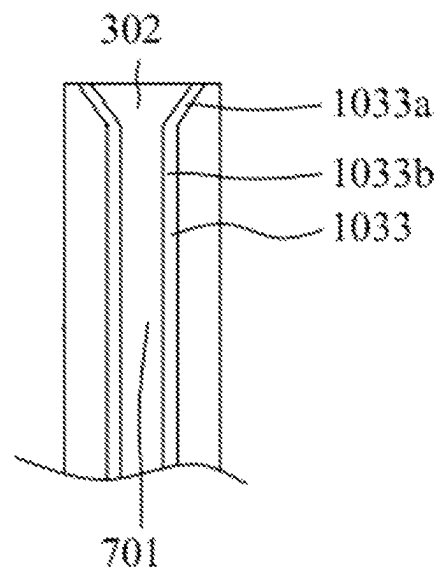


FIG. 5

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SPlicing DISPLAY DEVICE

FIELD OF DISCLOSURE

The present disclosure relates to the field of display technologies, in particular to a splicing display device.

BACKGROUND

At present, it is one of ways to realize seamless splicing of viewing angles of large-size display screens by arranging light-emitting diode strips correspondingly at a splicing position of two liquid crystal display panels which are connected and spliced. However, a flatness of the light-emitting diode strips fixed on the liquid crystal display panel directly affects a display performance of the large-size display screens.

SUMMARY OF DISCLOSURE

A purpose of the present disclosure is to provide a splicing display device to improve a flatness of a splicing position of non-display areas of two adjacent first display units where a second display unit is fixed to.

A splicing display device includes:

at least two first display units which are spliced and connected, where each of the first display units includes a non-display area; and a second display unit disposed on a light-emitting side of the first display units, and attached and fixed to a splicing position of the non-display areas of the two adjacent first display units.

In the splicing display device of some embodiments, the splicing display device further includes:

a vacuum suction component disposed in the non-display area of one of the first display units and arranged correspondingly to the second display unit, where the vacuum suction component is configured to attach the second display unit; and

a vacuuming unit disposed on a side of the one of the first display units away from the second display unit, where the vacuuming unit is connected to the vacuum suction component, and is configured to vacuum the vacuum suction component.

In the splicing display device of some embodiments, the vacuum suction component includes a suction hole, and the suction hole extends in a thickness direction of the first display units.

The splicing display device further includes a connecting pipe, one end of the connecting pipe is connected to the vacuuming unit, and another end of the connecting pipe is inserted into the suction hole.

In the splicing display device of some embodiments, each of the first display units further includes:

a first accommodating slot disposed in the non-display area of the first display unit, disposed on a surface of the first display unit close to the second display unit, and arranged correspondingly to the second display unit; and

a through hole correspondingly arranged and communicated with the first accommodating slot, and extending in a thickness direction of the first display unit.

The vacuum suction component includes a vacuum sucker and is disposed in the first accommodating slot.

The splicing display device further includes a connecting pipe, at least a portion of the connecting pipe is disposed in the through hole, one end of the connecting pipe is con-

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nected to the vacuum sucker, and another end of the connecting pipe is connected to the vacuuming unit.

In the splicing display device of some embodiments, the first display units are liquid crystal display modules, and the second display unit is a light emitting-diode light panel.

In the splicing display device of some embodiments, each of the first display units includes a liquid crystal display panel and a backlight module, the backlight module includes a backplane, the backplane includes a bottom plate and a side plate arranged along an edge of the bottom plate, the side plate of the backplane is arranged around the liquid crystal display panel, the side plate of the backplane is disposed in the non-display area of the first display unit, and the vacuum suction component is disposed in the side plate.

In the splicing display device of some embodiments, the splicing display device further includes:

a vacuum valve connected between the vacuum suction component and the vacuuming unit.

In the splicing display device of some embodiments, a group of the vacuum suction components is correspondingly disposed on the splicing position of the non-display area of one of the two adjacent first display units, the group of the vacuum suction components includes a plurality of the vacuum suction components, one of the vacuum valves is connected to the group of the vacuum suction components.

In the splicing display device of some embodiments, two groups of the vacuum suction components respectively disposed on the splicing position of the non-display areas of the two adjacent first display units are symmetrically arranged.

In the splicing display device of some embodiments, the plurality of the vacuum suction components of the group of the vacuum suction components are arranged side by side along an extending direction of a splicing seam of the splicing position.

In the splicing display device of some embodiments, each of the first display units includes:

a second accommodating slot disposed on a rear surface of the first display unit away from the second display unit.

The vacuum valve is detachably fixed in the second accommodating slot.

In the splicing display device of some embodiments, the splicing display device further includes:

a rear casing disposed on a back side of light-emitting surfaces of at least two of the first display units which are spliced, where the rear casing includes an accommodating cavity, and the vacuum valve and the vacuuming unit are arranged in the accommodating cavity of the rear casing.

The present disclosure provides the splicing display device. By attaching and fixing the second display unit to a splicing position of the non-display areas of the two adjacent first display units, a flatness of the splicing position the non-display areas of the two adjacent first display units where the second display unit is fixed to is improved, thereby improving a display performance of the splicing display device.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic cross-sectional view of a splicing display device according to an embodiment of the present disclosure.

FIG. 2 is a schematic plan view of a first display unit and a vacuum valve in the splicing display device shown in FIG. 1.

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FIG. 3 is a schematic cross-sectional view along a line A-A shown in FIG. 2.

FIG. 4 is a schematic cross-sectional view of a splicing display device according to another embodiment of the present disclosure.

FIG. 5 is a partially enlarged schematic view of the splicing display device shown in FIG. 4.

DETAILED DESCRIPTION

Technical solutions in the embodiments of the present disclosure will be clearly and completely described below with reference to the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only some, but not all, embodiments of the present disclosure. Based on the embodiments in the present disclosure, all other embodiments obtained by those skilled in the art without creative efforts shall fall within a protection scope of the present disclosure.

Referring to FIG. 1, FIG. 2, and FIG. 3, the present disclosure provides a splicing display device 100. The splicing display device 100 includes at least two first display units 10, a second display unit 20, a vacuum suction component 30, a vacuum valve 40, a vacuum unit 50, a rear casing 60, a first connecting pipe 701, and a second connecting pipe 702.

The first display units 10 are configured to display. Each of the first display units 10 includes a display area 10a and a non-display area 10b. The non-display area 10b is disposed at a periphery of the display area 10a and is arranged around the display area 10a.

The first display units 10 are liquid crystal display modules. Each of the first display units 10 includes a liquid crystal display panel 102 and a backlight module 103 disposed on a back side of a light-emitting surface of the liquid crystal display panel 102.

The backlight module 103 includes a backplane 1031, a light plate, a diffuser plate, an optical film, and the like. The backplane 1031 includes a bottom plate 1032 and four first side plates 1033. The bottom plate 1032 is rectangular. The four first side plates 1033 are arranged along four edges of the bottom plate 1032. The four first side plates 1033 and the bottom plate 1032 are surrounded to form a first accommodating cavity 1031a. The light plate, the diffuser plate, and the optical film etc. are arranged in the first accommodating cavity 1031a.

The liquid crystal display panel 102 is fixed on a first side plate 1033 of the backplane 1031. The first side plate 1033 is disposed around the liquid crystal display panel 102. A display area of the liquid crystal display panel 102 is the display area 10a of the first display unit 10. The non-display area 10b of the first display unit 10 includes a non-display area of the liquid crystal display panel 102. The non-display area 10b of the first display units 10 is provided with the first side plate 1033 of the backplane 1031.

It can be understood that the first display units 10 may also be micro light emitting diode display devices, sub-millimeter light emitting diode display devices, organic light emitting diode display devices or quantum dot display devices.

At least two first display units 10 may be spliced and connected along a same direction, or may be spliced and connected along two perpendicularly intersecting directions. Specifically, at least two first display units 10 are spliced and connected in the same direction. The two adjacent spliced first display units have a splicing position M. The splicing

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position M includes a splicing seam and the non-display areas 10b of the first display units 10.

In order to prevent the splicing seam of the splicing position M of the two adjacent spliced first display units and the non-display areas 10b of the first display units 10 from being viewed by human eyes, each second display unit 20 is disposed on the light-emitting side of the first display units, and is attached and fixed to the splicing position M of the non-display area 10b of the two adjacent first display units 10.

The second display unit 20 is a light emitting-diode light panel. The second display unit 20 includes a substrate and a plurality of inorganic light-emitting diodes arranged on the substrate. The substrate is a printed circuit board, and the inorganic light-emitting diodes can be micro light-emitting diodes or sub-millimeter light-emitting diodes.

The splicing display device 100 further includes a flexible printed circuit board 80 and a control circuit board 90. One end of the flexible printed circuit board 80 is connected to a rear surface of the light-emitting surface of the second display unit 20, and another end of the flexible printed circuit board 80 is connected to the control circuit board 90, so that the control circuit board 90 can control the second display unit 20 to emit light.

The splicing display device 100 also includes a middle frame 110. The middle frame 110 includes a frame body 1101 and four second side plates 1102 connected with the frame body 1101. The frame body 1101 is a rectangular frame body. The frame body 1101 is disposed on a side of the liquid crystal display panel 102 away from the backlight module 103 and is parallel to the liquid crystal display panel 102. The frame body 1101 is disposed in the non-display area of the splicing display device 100. An opening of the frame body 1101 overlaps with the display areas 10a of the at least two first display units 10 and the second display unit 20. The four second side plates 1102 are vertically connected to the frame body 1101. The four second side plates 1102 are disposed one-to-one with the four first side plates 1033 and are disposed at a periphery of the four first side plates 1033. The second side plate 1102 is detachably fixed on the first side plate 1033.

The rear casing 60 is disposed on a back side of the at least two spliced first display units 10. The rear casing 60 is detachably fixed to a rear surface of the first display units 10. The rear casing 60 includes a second accommodating cavity 60a. Specifically, the rear casing 60 is detachably fixed on the bottom plate 1032 of the backplane 1031.

In this embodiment, the splicing seam of the splicing position M and the non-display areas 10b of the first display units 10 are blocked by the second display unit 20. The second display unit 20 displays, which prevents the splicing seam of the splicing position M of the two adjacent spliced first display units and the non-display areas 10b of the first display units 10 from being viewed by the human eye. Moreover, for bonding the second display unit and the first display unit with an adhesive layer with a thickness of millimeters, unevenness and fixing problems are caused due to the uneven thickness of the adhesive layer. Compared with the above, in this embodiment, since each second display unit 20 is fixed to the non-display area 10b of the first display unit 10 by suction, the second display unit 20 can be fixed to the non-display area 10b of the first display unit 10 flatly, thereby improving the display performance of the splicing display device 100.

The vacuum suction component 30 is configured to attach the second display unit 20. The vacuum suction component

30 is disposed in the non-display area **10b** of the first display unit **10**, and is arranged corresponding to the second display unit **20**.

Referring to FIG. 2 and FIG. 3, a group of the vacuum suction components **30a** are correspondingly disposed in the splicing position M of the non-display areas **10b** of two adjacent first display units **10**. The group of the vacuum suction components **30a** includes a plurality of the vacuum suction components **30**.

The plurality of vacuum suction components **30** of the group of vacuum suction components **30a** are arranged side by side along an extending direction of the splicing seam. The plurality of vacuum suction components **30** of the group of vacuum suction components **30a** may also be arranged in a single row, two rows, or multiple rows. The plurality of vacuum suction components **30** of the group of vacuum suction components **30a** may also be arranged in a rectangular or other shape.

When the plurality of vacuum suction components **30** of the group of vacuum suction components **30a** are arranged side by side along the extension direction of the splicing seam, a distance between any two adjacent vacuum suction components **30** is equal in the extension direction of the splicing seam.

Two groups of the vacuum suction components **30a** respectively disposed on the splicing position M of the non-display areas **10b** of the two adjacent first display units **10** are symmetrically arranged. Suction forces of the two groups of the vacuum suction components **30a** to the second display unit **20** are the same, and the second display unit **20** can be fixed to the non-display areas **10b** of the two adjacent first display units **10** more smoothly.

The vacuum suction component **30** includes a suction hole **301**. The suction hole **301** is disposed on the first side plate **1033** of the backplane **1031** close to the splicing seam. The suction hole **301** extends in a thickness direction of the first display units **10** and extends through the first side plate **1033**.

The suction hole **301** is a cylindrical hole. It can be understood that the suction hole **301** may also be a quadrangular prism hole or a triangular prism hole.

The vacuuming unit **50** is configured to vacuum the vacuum suction component **30**. The vacuuming unit **50** is disposed on a side of the first display unit **10** away from the second display unit **20**. The vacuuming unit **50** is arranged in the second accommodating cavity **60a** of the rear casing **60**. The vacuuming unit **50** may be a vacuum compressor.

The vacuum valve **40** is configured to adjust a negative pressure condition in the vacuum suction component **30**. The vacuum valve **40** is connected between the vacuum suction component **30** and the vacuuming unit **50**. One vacuum valve **40** is connected to one group of the vacuum suction components **30a**.

Specifically, one ends of the plurality of first connecting pipes **701** are connected to the plurality of first ends of the vacuum valves **40** one-to-one. Another ends of the first connecting pipe **701** are inserted into the suction holes **301**. There is good sealing between the another end of the first connecting pipe **701** and the suction hole **301**, which is beneficial to maintain the negative pressure condition in the vacuum suction component **30**. One end of the second connecting pipe **702** is connected to the second end of the vacuum valve **40**. Another end of the second connecting pipe **702** is connected to the vacuuming unit **50**. Both the first connecting pipe **701** and the second connecting pipe **702** are plastic flexible pipes, which can be configured to allow gas to pass through.

As shown in FIG. 3, the first display unit **10** includes a second accommodating slot **1032c**. The second accommodating slot **1032c** is disposed on a rear surface of the first display unit **10** away from the second display unit **20**. The second accommodating slot **1032c** corresponds to the vacuum suction component **30**. The vacuum valve **40** is detachably fixed in the second accommodating slot **1032c**.

Specifically, the second accommodating slot **1032c** is disposed on a surface of the bottom plate **1032** of the backplane **1031** away from the liquid crystal display panel **102**, and the vacuum valve **40** is disposed in the second accommodating slot **1032c**.

It is understood that the vacuum valve **40** may also be disposed in the second accommodating cavity **60a** of the rear casing **60**.

In the splicing display device of this embodiment, the suction hole **301** is vacuumed by the vacuuming unit **50**, so that a pressure in the suction hole **301** is lower than an atmospheric pressure. Under an action of a pressure difference between the pressure in the suction hole **301** and the atmospheric pressure, the second display unit **20** is attached on the suction hole **301**, so that the second display unit **20** is attached and fixed to the non-display area **10b** of the first display unit **10**. Moreover, the vacuum valve **40** can adjust the negative pressure condition in the suction hole **301**, thereby adjusting a flatness of the non-display area **10b** of the first display unit **10** where the second display unit **20** is attached and fixed to.

Referring to FIG. 4 and FIG. 5, the splicing display device shown in FIG. 4 is basically similar to the splicing display device shown in FIG. 1, a difference is that the first display unit **10** further includes a first accommodating slot **1033a** and a through hole **1033b**. The first accommodating slot **1033a** is disposed on a surface of the first display unit **10** adjacent to the second display unit **20**. The first accommodating slot **1033a** is disposed in the non-display area **10b** of the first display unit **10** and is arranged corresponding to the second display unit **20**. The through hole **1033b** corresponds to the first accommodating slot **1033a** and they are communicated with each other. The through hole **1033b** extends in a thickness direction of the first display units **10**. The vacuum suction component **30** includes a vacuum sucker **302**. The vacuum sucker **302** is disposed in the first accommodating slot **1033a**. The first connecting pipe **701** is disposed in the through hole **1033b**. One end of the first connecting pipe **701** is connected to the vacuum sucker **302**. Another end of the first connecting pipe **701** is connected to the vacuum valve **40**. The second connecting pipe **702** connects the vacuum valve **40** and the vacuuming unit **50**. The second connecting pipe **702** is disposed in the second accommodating cavity **60a** of the rear casing **60**.

Specifically, the first accommodating slot **1033a** and the through hole **1033b** are disposed on the first side plate **1033** of the backplane **1031**. The first accommodating slot **1033a** is disposed on a surface of the first side plate **1033** close to the second display unit **20**. An area of a cross-section of the first accommodating slot **1033a** in a direction parallel to the bottom plate **1032** decreases from being close to the second display unit **20** to being away from the second display unit **20**. The first accommodating slot **1033a** is funnel-shaped. The through hole **1033b** extends in the thickness direction of the first display units **10**. The through hole **1033b** communicates with the first accommodating slot **1033a**. A shape of the vacuum sucker **302** is the same as that of the first accommodating slot **1033a**. The vacuum sucker **302** is disposed in the first accommodating slot **1033a**. One end of the first connecting pipe **701** is connected to vacuum sucker

302. Another end of the first connecting pipe 701 is connected to the vacuum valve 40. The second connecting pipe 702 connects the vacuum valve 40 and the vacuuming unit 50. The second connecting pipe 702 is disposed in the second accommodating cavity 60a of the rear casing 60.

In this embodiment, the vacuum sucker 302 is configured to attach the second display unit 20, which is beneficial to improve an attaching effect of the second display unit 20.

The descriptions of the above embodiments are only used to help understand the technical solutions and core ideas of the present disclosure. Those of ordinary skill in the art should understand that they can still make modifications to the technical solutions described in the foregoing embodiments, or perform equivalent replacements to some of the technical features. These modifications or replacements do not make the essence of the corresponding technical solutions deviate from the scope of the technical solutions of the various embodiments of the present disclosure.

What is claimed is:

1. A splicing display device, comprising:
 - at least two first display units which are spliced and connected, wherein each of the first display units comprises a non-display area, and the first display units are liquid crystal display modules; and
 - a second display unit disposed on a light-emitting side of the first display units, and attached and fixed to a splicing position of the non-display areas of the two adjacent first display units, wherein the second display unit comprises a substrate and a plurality of inorganic light-emitting diodes disposed on the substrate, and the inorganic light-emitting diodes are micro light-emitting diodes or sub-millimeter light-emitting diodes;
 - a vacuum suction component disposed in the non-display area of one of the first display units and arranged corresponding to the second display unit, wherein the vacuum suction component is configured to attach the second display unit.
2. The splicing display device according to claim 1, wherein further comprising:
 - a vacuuming unit disposed on a side of the one of the first display units away from the second display unit, wherein the vacuuming unit is connected to the vacuum suction component, and is configured to vacuum the vacuum suction component.
3. The splicing display device according to claim 2, wherein the vacuum suction component comprises a suction hole, and the suction hole extends in a thickness direction of the first display units; and
 - the splicing display device further comprises a connecting pipe, one end of the connecting pipe is connected to the vacuuming unit, and another end of the connecting pipe is inserted into the suction hole.
4. The splicing display device according to claim 2, wherein each of the first display units further comprises:
 - a first accommodating slot disposed in the non-display area of the first display unit, disposed on a surface of the first display unit close to the second display unit, and arranged corresponding to the second display unit; and
 - a through hole correspondingly arranged and communicated with the first accommodating slot, and extending in a thickness direction of the first display unit; wherein the vacuum suction component comprises a vacuum sucker and is disposed in the first accommodating slot; and
 - wherein the splicing display device further comprises a connecting pipe, at least a portion of the connecting

pipe is disposed in the through hole, one end of the connecting pipe is connected to the vacuum sucker, and another end of the connecting pipe is connected to the vacuuming unit.

5. The splicing display device according to claim 1, wherein each of the first display units comprises a liquid crystal display panel and a backlight module, the backlight module comprises a backplane, the backplane comprises a bottom plate and a side plate arranged along an edge of the bottom plate, the side plate of the backplane is arranged around the liquid crystal display panel, the side plate of the backplane is disposed in the non-display area of the first display unit, and the vacuum suction component is disposed in the side plate.

6. The splicing display device according to claim 2, further comprising: a vacuum valve connected between the vacuum suction component and the vacuuming unit.

7. The splicing display device according to claim 6, wherein a group of the vacuum suction components is correspondingly disposed on the splicing position of the non-display area of one of the two adjacent first display units, the group of the vacuum suction components comprises a plurality of the vacuum suction components, one of the vacuum valves is connected to the group of the vacuum suction components.

8. The splicing display device according to claim 7, wherein two groups of the vacuum suction components respectively disposed on the splicing position of the non-display areas of the two adjacent first display units are symmetrically arranged.

9. A splicing display device, comprising:

- at least two first display units which are spliced and connected, wherein each of the first display units comprises a non-display area; and

- a second display unit disposed on a light-emitting side of the first display units, and attached and fixed to a splicing position of the non-display areas of the two adjacent first display units;

- a vacuum suction component disposed in the non-display area of one of the first display units and arranged corresponding to the second display unit, wherein the vacuum suction component is configured to attach the second display unit.

10. The splicing display device according to claim 9, wherein further comprising:

- a vacuuming unit disposed on a side of the one of the first display units away from the second display unit, wherein the vacuuming unit is connected to the vacuum suction component, and is configured to vacuum the vacuum suction component.

11. The splicing display device according to claim 10, wherein the vacuum suction component comprises a suction hole, and the suction hole extends in a thickness direction of the first display units; and

- the splicing display device further comprises a connecting pipe, one end of the connecting pipe is connected to the vacuuming unit, and another end of the connecting pipe is inserted into the suction hole.

12. The splicing display device according to claim 10, wherein each of the first display units further comprises:

- a first accommodating slot disposed in the non-display area of the first display unit, disposed on a surface of the first display unit close to the second display unit, and arranged corresponding to the second display unit; and

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a through hole correspondingly arranged and communicated with the first accommodating slot, and extending in a thickness direction of the first display unit;

wherein the vacuum suction component comprises a vacuum sucker and is disposed in the first accommodating slot; and

wherein the splicing display device further comprises a connecting pipe, at least a portion of the connecting pipe is disposed in the through hole, one end of the connecting pipe is connected to the vacuum sucker, and another end of the connecting pipe is connected to the vacuuming unit.

13. The splicing display device according to claim **10**, wherein the first display units are liquid crystal display modules, and the second display unit is a light emitting-diode light panel.

14. The splicing display device according to claim **13**, wherein each of the first display units comprises a liquid crystal display panel and a backlight module, the backlight module comprises a backplane, the backplane comprises a bottom plate and a side plate arranged along an edge of the bottom plate, the side plate of the backplane is arranged around the liquid crystal display panel, the side plate of the backplane is disposed in the non-display area of the first display unit, and the vacuum suction component is disposed in the side plate.

15. The splicing display device according to claim **10**, further comprising: a vacuum valve connected between the vacuum suction component and the vacuuming unit.

16. The splicing display device according to claim **15**, wherein a group of the vacuum suction components is

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correspondingly disposed on the splicing position of the non-display area of one of the two adjacent first display units, the group of the vacuum suction components comprises a plurality of the vacuum suction components, one of the vacuum valves is connected to the group of the vacuum suction components.

17. The splicing display device according to claim **16**, wherein two groups of the vacuum suction components respectively disposed on the splicing position of the non-display areas of the two adjacent first display units are symmetrically arranged.

18. The splicing display device according to claim **16**, wherein the plurality of the vacuum suction components of the group of the vacuum suction components are arranged side by side along an extending direction of a splicing seam of the splicing position.

19. The splicing display device according to claim **15**, wherein each of the first display units comprises a second accommodating slot disposed on a rear surface of the first display unit away from the second display unit;

wherein the vacuum valve is detachably fixed in the second accommodating slot.

20. The splicing display device according to claim **15**, further comprising: a rear casing disposed on a back side of light-emitting surfaces of at least two of the first display units which are spliced, wherein the rear casing comprises an accommodating cavity, and the vacuum valve and the vacuuming unit are arranged in the accommodating cavity of the rear casing.

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